



Unpacking the Climate Signals of 2015: El Niño, Greenhouse Gases, and Extreme Weather

Abstract

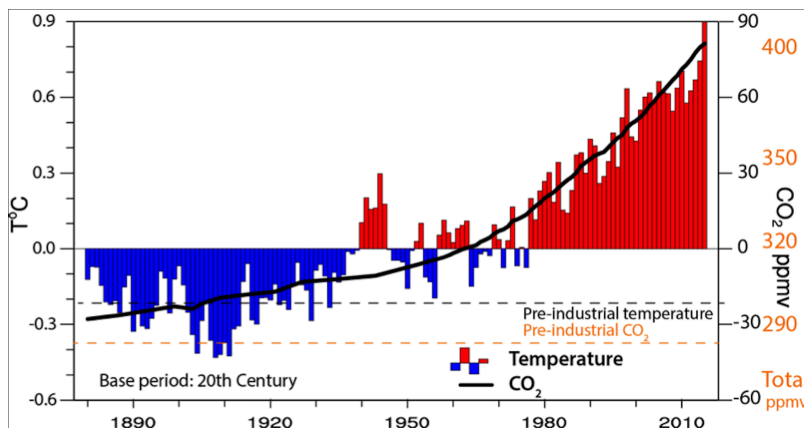
NASA and NOAA data showed that 2015 was the hottest year on record by far, beating 2014 by perhaps the largest margin ever. Climate scientist Kevin Trenberth explained how rising greenhouse gases and a very strong El Niño combined to produce a year of record heat waves, droughts, wildfires, floods, and storms around the world, offering a glimpse of what may be routine within about 15 years.

Introduction

Global mean surface temperature is the yardstick scientists use to track Earth's air temperature from year to year. In 2015 it set a new record, and the jump above 2014, the previous record-holder, was perhaps the largest on record. For temperature, 2015 was more than 1°C above pre-industrial levels. Carbon dioxide, measured at Mauna Loa in Hawaii, had risen from about 280 parts per million before industrialization to over 400. Trenberth argued that these data should dispel claims that global warming had stopped during the 1999 to 2013 hiatus. Instead, 2015 showed global warming alive and well, amplified by one of the strongest El Niños in the record.

How Do Scientists Study This?

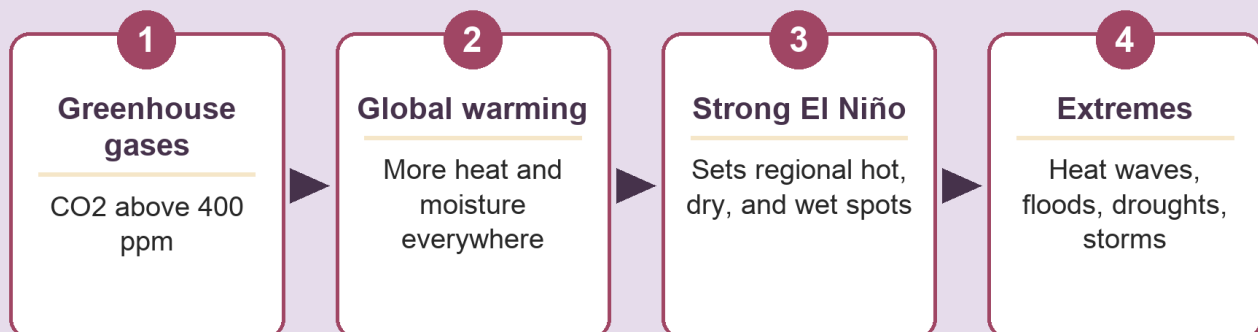
Trenberth and colleague John Fasullo plotted annual global temperature anomalies alongside carbon dioxide concentrations from NOAA, both compared with 20th-century averages and pre-industrial estimates. They compared 2015 with previous records and with climate model expectations. To understand the year's weather, Trenberth separated two influences: the long-term warming from greenhouse gases, and the regional patterns set by El Niño, which was only the third on record rated very strong. He then reviewed major weather events around the world and considered how each one fit the combined signal. He also applied basic physics: warmer air holds about 4% more water vapor for every 1°F rise.



FROM THE ARTICLE

Annual global temperature anomalies (bars) and carbon dioxide at Mauna Loa (line), relative to the 20th century, with pre-industrial values dashed. Kevin Trenberth/John Fasullo, data from NOAA.

How 2015 set records



Study the Process

Plot the records

Global temperatures and carbon dioxide levels were compared with pre-industrial and 20th-century values.

Separate the signals

Long-term greenhouse warming was distinguished from regional effects of El Niño.

Review the events

Heat waves, floods, droughts, and storms were examined for the combined fingerprint.

What Did They Discover?

El Niño could account for most of the difference between 2015 and 2014, but global warming made its effects more brutal. The Northern Hemisphere storm season broke records, driven by record numbers of category 4 and 5 Pacific typhoons. Deadly heat waves struck Eurasia: Madrid reached 104°F, Iran 115°F, and India 122°F, where about 2,300 people died. Warmer air fueled torrential rains: Missouri received almost three times its normal November-December precipitation, and Chennai flooded. Meanwhile, droughts and wildfires hit Indonesia, South Africa, Ethiopia, and western North America. Trenberth concluded that weather like 2015's will likely be routine in about 15 years, underscoring the importance of the Paris Agreement.

2015 by the numbers

1°C+

above pre-industrial global temperature

400+

ppm carbon dioxide, up from about 280

~15 yrs

until 2015-like weather may be routine

Floods in both the Northern and Southern Hemispheres at the same time, in the U.S. and Paraguay, are a characteristic fingerprint of El Niño patterns.

Look at the Evidence



FROM THE ARTICLE

Flooding in Chennai was driven by storms fueled by higher-than-average temperatures in the Bay of Bengal.
Photo: Anindito Mukherjee/Reuters.

Takeaways

Record heat

2015 was the hottest year on record, more than 1°C above pre-industrial levels.

Two forces combined

El Niño set regional patterns while global warming intensified them.

A glimpse ahead

Extremes like 2015's may become routine in about 15 years.

Key Terms

Temperature anomaly

How much a temperature differs from a long-term average, used to track warming over time.

Pre-industrial

The period before widespread fossil fuel use, often used as a baseline for measuring warming.

Parts per million (ppm)

A unit for measuring small concentrations, such as the amount of carbon dioxide in the air.

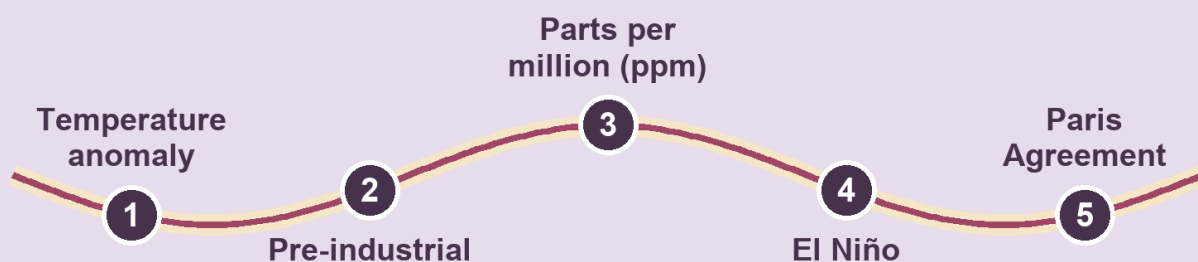
El Niño

A natural warming of the tropical Pacific that shifts weather patterns worldwide and raises global temperatures.

Paris Agreement

A 2015 international agreement that sets a framework for countries to slow climate change and prepare for its effects.

Vocabulary Path · Unpacking the Climate Signals of 2015



Activity

2015 Climate News Desk

Students report on one 2015 extreme weather event and explain how El Niño and global warming each contributed.

Materials: Event cards: India heat wave, Chennai floods, Missouri floods, Pacific typhoons, California wildfires; World map; Report template

1. Choose an event card and locate it on the world map.
2. Describe what happened, using numbers from the lesson.
3. Explain El Niño's role in where the event happened.
4. Explain global warming's role in how severe it was, using the 4% per °F rule if relevant.
5. Present a 60-second news report, ending with what the event suggests about the future.

References

Trenberth, K. (2016). The hottest year on record signals that global warming is alive and well. The Conversation. <https://doi.org/10.64628/AAI.5g5msartv>

Also published in French by The Conversation as "2015 aura bien été l'année la plus chaude jamais enregistrée" (article 53623); both editions are covered by this lesson.